

Title: Explainable Machine Learning Model for Distillery Effluent Treatment and Management

Introduction/Background:

Industrial effluent management remains a significant environmental challenge, particularly in developing countries such as Kenya, where rapid industrialization has increased the risks of water pollution and public health hazards. Distillery industries generate large volumes of wastewater characterized by high Chemical Oxygen Demand (COD), Biological Oxygen Demand (BOD), total dissolved solids, and toxic organic compounds. Conventional monitoring approaches are largely manual, reactive, and dependent on periodic laboratory testing, while many machine learning models operate as black-box systems that lack transparency and stakeholder trust.

Objective:

My study aims to develop an explainable machine learning model for effective distillery effluent treatment and management by integrating predictive analytics with interpretable artificial intelligence techniques.

Methodology:

The research adopts a quantitative approach guided by Design Science Research Methodology (DSRM). Secondary data obtained from publicly available wastewater datasets, complemented by validated synthetic data, used to train and evaluate predictive models. Environmental parameters such as pH, COD, BOD, temperature, and total suspended solids serve as input features. Machine learning algorithms, including Random Forest and Gradient Boosting, are integrated with SHapley Additive Explanations (SHAP) to enhance model interpretability. Model performance assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

Findings:

The study is expected to demonstrate that explainable machine learning models can achieve high predictive accuracy while providing transparent insights into the environmental and operational factors influencing effluent treatment performance. The integration of SHAP is anticipated to improve stakeholder trust, regulatory compliance, and decision-making capabilities.

Conclusion:

The proposed framework offers a transparent and data-driven approach to industrial wastewater management, contributing to sustainable environmental practices and providing a replicable model for other environmental monitoring applications in Kenya and beyond.

Keywords: Explainable Artificial Intelligence, Machine Learning, Distillery Effluent, Wastewater Treatment, SHAP, Environmental Monitoring, Kenya.